A three-level hierarchical composition rule for joint sampling, and its induction to N levels

COMPASS working note (derivation only, no code changes)

12 October 2026

Abstract

We derive the composition rule behind `gauss_hierarchical` and `gauss_jacobian` for a three-level hierarchy. The model has a top-level global g, mid-level (group) parameters h_k, and local parameters ℓ_{kj}, one per observation. The rule returns the score of the diffused tall-data posterior over the *whole* latent vector $(g, h_{1:K}, \ell_{1:K,\cdot})$, i.e. a composed score for the top-level globals, the mid-level globals and the locals. It can therefore drive any reverse-diffusion sampler to draw *joint* posterior samples, exactly as `gauss_hierarchical` does for $(g, \ell_{1:n})$. It is assembled only from the single-observation network's scores on root-to-leaf paths (g, h_k, ℓ_{kj}), Gaussian backward precisions on those paths (from a pilot run or from the network Jacobian), and the diffused prior on g and on each (g, h_k). The approximation is the same one used by `gauss_hierarchical`: every backward kernel is replaced by a Gaussian. Everything else is exact algebra. We then extend the derivation from three to N levels by induction. The N-level rule is one upward pass of merge-and-eliminate steps, one per level, followed by a downward cascade of Gaussian conditional-mean corrections. It costs time linear in the number of nodes and reduces to `gauss_hierarchical` for $N = 2$. A NumPy check on linear-Gaussian hierarchies reproduces the exact composed score to $\leq 4 \times 10^{-15}$ for $N = 2, \ldots, 5$. A probability-flow sampler driven by the three-level composed score recovers the exact joint posterior, including all cross-level correlations, to Monte Carlo noise.

Contents

1 Setting and notation **2**

2 Step 1: the exact three-level factorisation **2**

3 Step 2: the composed score is a backward-kernel mean **3**

4 Step 3: Gaussian factors give a nested arrowhead system **3**

5 Step 4: elimination, level by level **4**

5.1 Pass 1: eliminate the locals 4

5.2 Pass 2: merge each group, then eliminate the mid-level globals 5

5.3 Pass 3: the top-level globals 5

5.4 The three-level rule 6

6 Properties of the three-level rule **6**

7 Joint sampling with the three-level score **7**

8 The prior quantities at the mid level **8**

9 From three to N levels **8**
9.1 Nested N-level model 8
9.2 Step 1 for N levels 9
9.3 Steps 2–3 for N levels 9
9.4 Step 4 for N levels: induction over the levels 9
9.5 Numerical confirmation 11

10 Summary **11**

1 Setting and notation

Model.

$$p(\theta, \mathbf{x}) = p(g) \prod_{k=1}^{K} \Big[p(h_k \mid g) \prod_{j=1}^{n_k} p(\ell_{kj} \mid g, h_k)\, p(x_{kj} \mid g, h_k, \ell_{kj}) \Big], \tag{1}$$

with $g \in \mathbb{R}^{d_g}$ (top-level globals), $h_k \in \mathbb{R}^{d_h}$ (mid-level globals of group k), $\ell_{kj} \in \mathbb{R}^{d_\ell}$ (locals of observation kj), and $N_{\text{obs}} = \sum_k n_k$ observations. All conditional priors may depend on every ancestor. Example: g is a population mean and scale, h_k a school effect, ℓ_{kj} a pupil effect, and x_{kj} a pupil's data. Two-level partial pooling is the case $d_h = 0$.

Paths and the training unit. The latent vector is $\theta = (g, h_1, \ldots, h_K, \ell_{11}, \ldots, \ell_{Kn_K})$, of dimension $d_g + Kd_h + N_{\text{obs}} d_\ell$. Observation kj sees the root-to-leaf *path* $\vartheta_{kj} = (g, h_k, \ell_{kj}) \in \mathbb{R}^D$, $D = d_g + d_h + d_\ell$. The group sees $a_k = (g, h_k) \in \mathbb{R}^{d_a}$, $d_a = d_g + d_h$. The selection matrices satisfy $E_{kj}\theta = \vartheta_{kj}$, $E_{a_k}\theta = a_k$ and $E_g\theta = g$.

The score network is trained exactly as now, on single-observation simulations, but now with three latent tiers: draw $g \sim p(g)$, $h_k \sim p(h_k \mid g)$, $\ell_{kj} \sim p(\ell_{kj} \mid g, h_k)$ and $x_{kj} \sim p(x_{kj} \mid \vartheta_{kj})$. It returns the joint path score

$$s_{kj}(y_{kj}) = \nabla_{y_{kj}} \log p_t(y_{kj} \mid x_{kj}) = (s^g_{kj},\, s^h_{kj},\, s^\ell_{kj}), \qquad y_{kj} = (g_t, h_{k,t}, \ell_{kj,t}). \tag{2}$$

We also write $s^a_{kj} = (s^g_{kj}, s^h_{kj})$ for its (g, h) part.

Diffusion. Fix t and write $\lambda = \lambda_t$ (variance-exploding convention, as in the paper; VP is identical in $y = x/\alpha_t$ coordinates). The noisy state is $y = (g_t, h_{1,t}, \ldots, \ell_{Kn_K,t})$, and $\varphi_\lambda(z) = \mathcal{N}(z; 0, \lambda^2 \mathbf{I})$ factorises over coordinates. Backward kernels are written π_t; the prior ones carry a superscript 0.

Prior quantities.

- top prior on g: diffused score $s_0(g_t) = \nabla \log p_t(g_t)$ and Gaussian backward precision $P_0 \in \mathbb{R}^{d_g \times d_g}$;
- group prior on a_k: the marginal $p(g, h_k) = p(g)\, p(h_k \mid g)$, with diffused score $s_{0,k}(g_t, h_{k,t}) = (s^g_{0,k}, s^h_{0,k})$ and backward precision $P_{0,k} \in \mathbb{R}^{d_a \times d_a}$.

In the exchangeable case all $p(g, h_k)$ are the same distribution, so one prior model evaluated at K different points suffices.

2 Step 1: the exact three-level factorisation

Proposition 1.

$$p(\theta \mid \mathbf{x}) \;\propto\; p(g)^{1-K} \prod_{k=1}^{K} p(g, h_k)^{1-n_k} \prod_{k=1}^{K} \prod_{j=1}^{n_k} p(\vartheta_{kj} \mid x_{kj}). \tag{3}$$

Proof. By Bayes' rule, $p(\vartheta_{kj} \mid x_{kj}) = p(g)\, p(h_k \mid g)\, p(\ell_{kj} \mid g, h_k)\, p(x_{kj} \mid \vartheta_{kj})/p(x_{kj})$. Take the product over all kj and collect the exponent of each prior factor:

- $p(\ell_{kj} \mid g, h_k)$ and the likelihood $p(x_{kj} \mid \vartheta_{kj})$ appear once;
- $p(h_k \mid g)$ appears n_k times, and $p(g)$ appears $N_{\text{obs}} = \sum_k n_k$ times.

The correction $\prod_k p(g, h_k)^{1-n_k} = \prod_k p(g)^{1-n_k} p(h_k \mid g)^{1-n_k}$ brings each $p(h_k \mid g)$ to exponent 1 and $p(g)$ to $N_{\text{obs}} + \sum_k (1 - n_k) = K$. The remaining $p(g)^{1-K}$ brings $p(g)$ to 1. What is left is $p(\theta, \mathbf{x})/\prod p(x_{kj})$, which is proportional in θ to the posterior. □

Compared with two levels, there is *one correction per internal node*: $p(g)^{1-K}$ at the root and $p(g, h_k)^{1-n_k}$ at each group. The group correction is on the *joint* prior of the group path (g, h_k), not on $p(h_k \mid g)$ alone. This matters after diffusion, because the noise acts on g and h_k jointly.

Proposition 2 (Backward-kernel factorisation)**.** *For every t, keeping y and $\mathbf{x}$ fixed and reading proportionality in θ_0,*

$$\pi_t(\theta_0 \mid y, \mathbf{x}) \;\propto\; \pi_t^0(g_0 \mid g_t)^{1-K} \prod_k \pi_t^0(a_{k,0} \mid a_{k,t})^{1-n_k} \prod_{k,j} \pi_t(\vartheta_{kj,0} \mid y_{kj}, x_{kj}). \tag{4}$$

Proof. $\pi_t(\theta_0 \mid y, \mathbf{x}) \propto \varphi_\lambda(y - \theta_0)\, p(\theta_0 \mid \mathbf{x})$. Factorise the forward kernel coordinate-wise and split each coordinate's copy into as many powers as there are factors containing it. Each ℓ_{kj} is contained only in factor kj, so it gets exponent 1. Each h_k gets $(1 - n_k) + n_k = 1$, and g gets $(1 - K) + \sum_k (1 - n_k) + \sum_k n_k = 1$. Hence

$$\varphi_\lambda(y - \theta_0) = \varphi_\lambda(g_t - g_0)^{1-K} \prod_k \varphi_\lambda(a_{k,t} - a_{k,0})^{1-n_k} \prod_{k,j} \varphi_\lambda(y_{kj} - \vartheta_{kj,0}).$$

Multiply by (3) and pair each noise factor with the density factor of the same scope. Every pair is a backward kernel times a normaliser ($p_t(g_t)^{1-K}$, $p_t(a_{k,t})^{1-n_k}$, $p_t(y_{kj} \mid x_{kj})$) that does not depend on θ_0. □

3 Step 2: the composed score is a backward-kernel mean

Tweedie's formula holds for any distribution with a finite first moment, so the composed score of the full latent vector is

$$u(y) := \nabla_y \log p_t(y \mid \mathbf{x}) = \frac{\mu(y) - y}{\lambda^2}, \qquad \mu(y) = \mathbb{E}[\theta_0 \mid y, \mathbf{x}]. \tag{5}$$

u has one block per latent: u^g, u^{h_k} and $u^{\ell_{kj}}$. Applied to the factors, Tweedie gives their means in terms of the available scores:

$$m_{kj} = y_{kj} + \lambda^2 s_{kj}, \qquad m_{0,k} = a_{k,t} + \lambda^2 s_{0,k}, \qquad m_0 = g_t + \lambda^2 s_0. \tag{6}$$

4 Step 3: Gaussian factors give a nested arrowhead system

The single approximation. Replace every factor in (4) by a Gaussian with the correct mean (6): precision P_{kj} for $\pi_t(\vartheta_{kj,0} \mid \cdot)$, $P_{0,k}$ for $\pi_t^0(a_{k,0} \mid \cdot)$ and P_0 for $\pi_t^0(g_0 \mid \cdot)$. The paper's Step 3 then goes through verbatim, with weights $1 - K$, $1 - n_k$ and 1. The surrogate is $\exp(-\frac{1}{2}\theta_0^\top \Lambda \theta_0 + \theta_0^\top b)$, and since each mean and its state share a precision, y drops out:

$$\Lambda u = r, \qquad \begin{aligned} \Lambda &= (1 - K)\, E_g^\top P_0 E_g + \sum_k (1 - n_k)\, E_{a_k}^\top P_{0,k} E_{a_k} + \sum_{k,j} E_{kj}^\top P_{kj} E_{kj}, \\ r &= (1 - K)\, E_g^\top P_0 s_0 + \sum_k (1 - n_k)\, E_{a_k}^\top P_{0,k} s_{0,k} + \sum_{k,j} E_{kj}^\top P_{kj} s_{kj}. \end{aligned} \tag{7}$$

Blocks. Partition each path precision and each group-prior precision as

$$P_{kj} = \begin{pmatrix} P^{gg}_{kj} & P^{gh}_{kj} & P^{g\ell}_{kj} \\ P^{hg}_{kj} & P^{hh}_{kj} & P^{h\ell}_{kj} \\ P^{\ell g}_{kj} & P^{\ell h}_{kj} & P^{\ell\ell}_{kj} \end{pmatrix}, \qquad P_{0,k} = \begin{pmatrix} P^{gg}_{0,k} & P^{gh}_{0,k} \\ P^{hg}_{0,k} & P^{hh}_{0,k} \end{pmatrix}.$$

The non-zero blocks of Λ are

$$\begin{aligned} \Lambda_{gg} &= (1-K)P_0 + \sum_k (1-n_k)P^{gg}_{0,k} + \sum_{k,j} P^{gg}_{kj}, \quad \Lambda_{gh_k} = (1-n_k)P^{gh}_{0,k} + \sum_j P^{gh}_{kj}, \\ \Lambda_{h_k h_k} &= (1-n_k)P^{hh}_{0,k} + \sum_j P^{hh}_{kj}, \qquad \Lambda_{g\ell_{kj}} = P^{g\ell}_{kj}, \quad \Lambda_{h_k \ell_{kj}} = P^{h\ell}_{kj}, \quad \Lambda_{\ell_{kj}\ell_{kj}} = P^{\ell\ell}_{kj}, \end{aligned} \tag{8}$$

with transposes below the diagonal. All other blocks are zero: h_k–$h_{k'}$, ℓ–ℓ, and h_k–$\ell_{k'j}$ for $k' \neq k$. Ordered as $(g;\, h_1, \ell_{1\cdot};\, \ldots;\, h_K, \ell_{K\cdot})$, Λ is a *nested arrowhead*: an outer arrowhead whose diagonal blocks are themselves arrowheads,

$$\Lambda = \begin{pmatrix} \Lambda_{gg} & \mathcal{B}_1 & \cdots & \mathcal{B}_K \\ \mathcal{B}_1^\top & \mathcal{A}_1 & & \\ \vdots & & \ddots & \\ \mathcal{B}_K^\top & & & \mathcal{A}_K \end{pmatrix}, \qquad \mathcal{A}_k = \begin{pmatrix} \Lambda_{h_k h_k} & P^{h\ell}_{k1} & \cdots & P^{h\ell}_{kn_k} \\ P^{\ell h}_{k1} & P^{\ell\ell}_{k1} & & \\ \vdots & & \ddots & \\ P^{\ell h}_{kn_k} & & & P^{\ell\ell}_{kn_k} \end{pmatrix}.$$

The right-hand side blocks are, with $[\cdot]^g$ etc. denoting sub-vectors,

$$\begin{aligned} r^g &= (1-K)P_0 s_0 + \sum_k (1-n_k)[P_{0,k}s_{0,k}]^g + \sum_{k,j}[P_{kj}s_{kj}]^g, \\ r^{h_k} &= (1-n_k)[P_{0,k}s_{0,k}]^h + \sum_j [P_{kj}s_{kj}]^h, \qquad r^{\ell_{kj}} = [P_{kj}s_{kj}]^\ell. \end{aligned} \tag{9}$$

5 Step 4: elimination, level by level

We use the two block identities of the paper (§3.4) for a positive-definite $P = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right)$ with $\Sigma = P^{-1}$: $S := A - BC^{-1}B^\top = (\Sigma_{11})^{-1}$ and $-C^{-1}B^\top = \Sigma_{21}\Sigma_{11}^{-1}$.

5.1 Pass 1: eliminate the locals

Split each path precision with respect to (a_k, ℓ_{kj}):

$$P_{kj} = \begin{pmatrix} A_{kj} & B_{kj} \\ B_{kj}^\top & C_{kj} \end{pmatrix}, \quad A_{kj} = \begin{pmatrix} P^{gg}_{kj} & P^{gh}_{kj} \\ P^{hg}_{kj} & P^{hh}_{kj} \end{pmatrix}, \; B_{kj} = \begin{pmatrix} P^{g\ell}_{kj} \\ P^{h\ell}_{kj} \end{pmatrix}, \; C_{kj} = P^{\ell\ell}_{kj}.$$

Local row. By (8) and (9), the ℓ_{kj} row of $\Lambda u = r$ reads $B_{kj}^\top u^{a_k} + C_{kj} u^{\ell_{kj}} = B_{kj}^\top s^a_{kj} + C_{kj} s^\ell_{kj}$, so

$$u^{\ell_{kj}} = s^\ell_{kj} + K_{kj}(u^{a_k} - s^a_{kj}), \qquad K_{kj} := -C_{kj}^{-1}B_{kj}^\top = [\Sigma_{t,kj}]_{\ell a}[\Sigma_{t,kj}]_{aa}^{-1} \in \mathbb{R}^{d_\ell \times d_a}. \tag{10}$$

Splitting $K_{kj} = (K^g_{kj}, K^h_{kj})$ by columns, the local correction is $K^g_{kj}(u^g - s^g_{kj}) + K^h_{kj}(u^{h_k} - s^h_{kj})$. The local moves in response to how far the composed top-level *and* mid-level scores have moved away from observation kj's own opinion.

Effect on the (g, h_k) rows (first cancellation). Factor kj contributes $A_{kj}u^{a_k} + B_{kj}u^{\ell_{kj}}$ to the left-hand side of the a_k rows and $A_{kj}s^a_{kj} + B_{kj}s^\ell_{kj}$ to the right-hand side. Substituting (10),

$$A_{kj}u^{a_k} + B_{kj}s^\ell_{kj} - B_{kj}C^{-1}_{kj}B^\top_{kj}(u^{a_k} - s^a_{kj}) \;=\; A_{kj}s^a_{kj} + B_{kj}s^\ell_{kj}.$$

$B_{kj}s^\ell_{kj}$ cancels. The local score drops out of every remaining equation, and factor kj contributes $S_{kj}u^{a_k}$ on the left and $S_{kj}s^a_{kj}$ on the right, with

$$S_{kj} := A_{kj} - B_{kj}C^{-1}_{kj}B^\top_{kj} = \left([\Sigma_{t,kj}]_{aa}\right)^{-1} \in \mathbb{R}^{d_a \times d_a}, \tag{11}$$

the precision of the (g, h)-marginal of observation kj's Gaussian backward kernel. After pass 1, observation kj is the *message* (S_{kj}, s^a_{kj}) on (g, h_k). It carries the network's *conditional* (g, h) score at the current $\ell_{kj,t}$, weighted by the *marginal* precision. This is the two-level mechanism, applied with "global" $= (g, h_k)$.

5.2 Pass 2: merge each group, then eliminate the mid-level globals

Merge. After pass 1, the only factors that involve h_k are the n_k messages and the group prior, all with scope a_k. Their sum is

$$\boxed{Q_k := \sum_{j=1}^{n_k} S_{kj} + (1 - n_k)P_{0,k}, \qquad q_k := \sum_{j=1}^{n_k} S_{kj}\, s^a_{kj} + (1 - n_k)P_{0,k}\, s_{0,k}, \qquad \sigma_k := Q_k^{-1}q_k.} \tag{12}$$

$\sigma_k = (\sigma^g_k, \sigma^h_k) \in \mathbb{R}^{d_a}$ is *literally the two-level* `gauss_hierarchical` *rule applied to group k alone*, with global (g, h_k), locals $\ell_{k\cdot}$ and prior $p(g, h_k)$. It is the composed score of the *group* tall posterior $p(g, h_k, \ell_{k\cdot} \mid x_{k\cdot})$ on its (g, h) block. Q_k^{-1} is the covariance of that group's composed Gaussian backward kernel on (g, h_k), with its locals integrated out. In information form,

$$Q_k = P_{0,k} + \sum_{j=1}^{n_k} (S_{kj} - P_{0,k}) =: P_{0,k} + \sum_j I_{kj}, \tag{13}$$

so I_{kj} is the information observation kj carries about (g, h_k).

Mid-level row. Split $Q_k = \begin{pmatrix} Q^{gg}_k & Q^{gh}_k \\ Q^{hg}_k & Q^{hh}_k \end{pmatrix}$ and likewise $q_k = (q^g_k, q^h_k) = Q_k\sigma_k$. After pass 1, the h_k row of the system reads $Q^{hg}_k u^g + Q^{hh}_k u^{h_k} = q^h_k = Q^{hg}_k\sigma^g_k + Q^{hh}_k\sigma^h_k$, so

$$u^{h_k} = \sigma^h_k + G_k(u^g - \sigma^g_k), \qquad G_k := -(Q^{hh}_k)^{-1}Q^{hg}_k = [Q_k^{-1}]_{hg}\,[Q_k^{-1}]^{-1}_{gg} \in \mathbb{R}^{d_h \times d_g}. \tag{14}$$

Effect on the g row (second cancellation). Group k contributes $Q^{gg}_k u^g + Q^{gh}_k u^{h_k}$ on the left and $q^g_k = Q^{gg}_k\sigma^g_k + Q^{gh}_k\sigma^h_k$ on the right. Substituting (14), the $Q^{gh}_k\sigma^h_k$ terms cancel, and the group contributes $T_k u^g$ on the left and $T_k\sigma^g_k$ on the right, with

$$T_k := Q^{gg}_k - Q^{gh}_k(Q^{hh}_k)^{-1}Q^{hg}_k = \left([Q_k^{-1}]_{gg}\right)^{-1} \in \mathbb{R}^{d_g \times d_g}. \tag{15}$$

After pass 2, group k is the message (T_k, σ^g_k) on g: *a group behaves like a single observation* with score σ^g_k and precision T_k.

5.3 Pass 3: the top-level globals

The remaining g row is $\left[\sum_k T_k + (1 - K)P_0\right]u^g = \sum_k T_k\sigma^g_k + (1 - K)P_0 s_0$.

5.4 The three-level rule

Three-level composed score (all latents)

Given the network path scores s_{kj}, path precisions P_{kj}, and prior quantities (s_0, P_0) and $(s_{0,k}, P_{0,k})$:

$$\text{(pass 1)} \quad S_{kj} = A_{kj} - B_{kj}C_{kj}^{-1}B_{kj}^{\top}, \qquad K_{kj} = -C_{kj}^{-1}B_{kj}^{\top}, \tag{16}$$

$$\text{(pass 2)} \quad Q_k = \textstyle\sum_j S_{kj} + (1-n_k)P_{0,k}, \quad \sigma_k = Q_k^{-1}\big[\textstyle\sum_j S_{kj}s_{kj}^a + (1-n_k)P_{0,k}s_{0,k}\big],$$

$$T_k = Q_k^{gg} - Q_k^{gh}(Q_k^{hh})^{-1}Q_k^{hg}, \qquad G_k = -(Q_k^{hh})^{-1}Q_k^{hg}, \tag{17}$$

$$\text{(top)} \quad u^g = \Big[\textstyle\sum_{k=1}^K T_k + (1-K)P_0\Big]^{-1}\Big[\textstyle\sum_{k=1}^K T_k\,\sigma_k^g + (1-K)P_0\,s_0(g_t)\Big], \tag{18}$$

$$\text{(mid)} \quad u^{h_k} = \sigma_k^h + G_k(u^g - \sigma_k^g), \tag{19}$$

$$\text{(local)} \quad u^{\ell_{kj}} = s_{kj}^{\ell} + K_{kj}^g(u^g - s_{kj}^g) + K_{kj}^h(u^{h_k} - s_{kj}^h). \tag{20}$$

The *structure* of the rule is: compose upwards (locals → groups → top), then correct downwards (top → mid → locals). Its parts are:

- (18) is the two-level `gauss_hierarchical` global rule, with the *groups* as observations;
- each group score σ_k is the two-level rule over its own observations;
- (19) and (20) are two nested copies of the two-level local rule.

Denoiser reading. Multiply (19) and (20) by λ^2 and add the state. With the composed denoiser $\hat{\theta} = y + \lambda^2 u$, the group denoiser $\hat{a}_k = a_{k,t} + \lambda^2\sigma_k$ and the network denoiser $\hat{\vartheta}_{kj} = y_{kj} + \lambda^2 s_{kj}$:

$$\hat{h}_k = \hat{a}_k^h + G_k(\hat{g} - \hat{a}_k^g), \qquad \hat{\ell}_{kj} = \hat{\vartheta}_{kj}^{\ell} + K_{kj}(\hat{a}_k^{\text{comp}} - \hat{\vartheta}_{kj}^a), \quad \hat{a}_k^{\text{comp}} = (\hat{g}, \hat{h}_k).$$

These are Gaussian conditional-mean updates cascading down the tree. Once all data have revised the estimate of g, each group's h estimate shifts by the regression G_k of h on g under that group's kernel. Each local then shifts by the regression K_{kj} of ℓ on (g, h) under its own kernel.

Cost. N_{obs} inversions of $d_\ell \times d_\ell$ blocks, K inversions of $d_h \times d_h$ blocks (and of Q_k, $d_a \times d_a$), and one $d_g \times d_g$ solve. This is linear in N_{obs} and K, with no dense $(d_g + Kd_h + N_{\text{obs}}d_\ell)$-dimensional solve.

Remark 1 (Numerically preferred form). Forming σ_k needs Q_k to be invertible; the elimination only needs $Q_k^{hh} \succ 0$. Equivalently, use the information vector $\tau_k = q_k^g - Q_k^{gh}(Q_k^{hh})^{-1}q_k^h$ $(= T_k\sigma_k^g)$ in (18) and $u^{h_k} = (Q_k^{hh})^{-1}(q_k^h - Q_k^{hg}u^g)$ in place of (19).

6 Properties of the three-level rule

Proposition 3 (Well-posedness)**.** *$\Lambda \succ 0$ if and only if every pivot of the elimination is positive definite: $C_{kj} \succ 0$ (automatic, since $P_{kj} \succ 0$), $Q_k^{hh} \succ 0$, and $\Lambda_T := \sum_k T_k + (1-K)P_0 \succ 0$. A sufficient condition is that every information increment is PSD:*

$$I_{kj} = S_{kj} - P_{0,k} \succeq 0 \quad \textit{and} \quad J_k := T_k - P_0 \succeq 0, \tag{21}$$

since then $Q_k \succeq P_{0,k} \succ 0$ and $\Lambda_T = P_0 + \sum_k J_k \succeq P_0 \succ 0$.

Proof. The three passes form a block $LDL^{\top}$ factorisation of Λ with those pivots. The information forms follow from $\sum_j S_{kj} + (1-n_k)P_{0,k} = P_{0,k} + \sum_j (S_{kj} - P_{0,k})$, and similarly for Λ_T. □

In the linear-Gaussian case both increments are exactly PSD. S_{kj} is the precision of (g, h) given (y_{kj}, x_{kj}), and $P_{0,k}$ that given $a_{k,t}$ alone. T_k is the precision of g given all of group k's states and data, and P_0 that given g_t alone. Gaussian conditioning on more never increases covariance. The existing PSD projection of I_j therefore generalises to projecting both I_{kj} and J_k.

Exactness. If $p(g)$, $p(g, h_k)$ and all $p(\vartheta_{kj} \mid x_{kj})$ are Gaussian (a linear-Gaussian three-level model), every backward kernel is Gaussian. With exact path scores and covariances, the rule gives the exact composed score at every t.

Consistency checks. Each follows by comparing (Λ, r).

1. $N_{\text{obs}} = 1$: all corrections have exponent 0, and $u = s_{11}$.
2. $n_k = 1$ for all k: the group priors drop out, $Q_k = S_{k1}$ and $\sigma_k = s^a_{k1}$. This is `gauss_hierarchical` with global g and local block (h_k, ℓ_{k1}).
3. $K = 1$: the top prior drops out and $u^g = \sigma^g_1$. This is `gauss_hierarchical` with global (g, h_1) and prior $p(g, h_1)$.
4. $d_h = 0$: $P_{0,k} = P_0$ and $T_k = Q_k$. Then (18) becomes $\big[\sum_{kj} S_{kj} + (1 - N_{\text{obs}})P_0\big]^{-1} \big[\sum_{kj} S_{kj} s^g_{kj} + (1 - N_{\text{obs}})P_0 s_0\big]$, which is `gauss_hierarchical` over all N_{obs} observations.
5. No coupling of the locals ($B_{kj} = 0$): $K_{kj} = 0$ and $u^{\ell_{kj}} = s^\ell_{kj}$. No coupling of the mid level ($Q^{gh}_k = 0$): $G_k = 0$ and $T_k = Q^{gg}_k$.

7 Joint sampling with the three-level score

Yes: the rule gives a score for every latent. $u(y)$ from (18)–(20) is (up to the Gaussian-kernel approximation) $\nabla_y \log p_t(y \mid \mathbf{x})$ for the *full* vector $y = (g_t, h_{1:K,t}, \ell_{\cdot,t})$. Plugged into any reverse-time sampler (Euler/DPM predictor, churn, Langevin correctors or the probability-flow ODE), it moves all coordinates together. At $t \to 0$ the sampler returns draws from the joint posterior $p(g, h_{1:K}, \ell_{1:K,\cdot} \mid \mathbf{x})$, with consistent cross-level dependence: g–h, h–ℓ and g–ℓ. No Gibbs-style alternation between levels is needed. This is the same situation as `gauss_hierarchical`, whose u^g and u^{ℓ_j} drive one joint sampler.

One sampler step. At (t, y):

1. For every observation, gather the path state $y_{kj} = (g_t, h_{k,t}, \ell_{kj,t})$ and evaluate the network once (batched over all N_{obs} paths) to get s_{kj}.
2. Path precisions P_{kj}:

`gauss_hierarchical`: $P_{kj} = \hat{\Sigma}^{-1}_{0,kj} + \lambda^{-2}\mathbf{I}_D$, with $\hat{\Sigma}_{0,kj}$ the pilot covariance of the network's single-observation draws of the *whole path* (g, h_k, ℓ_{kj}) given x_{kj}.

`gauss_jacobian`: $P_{kj} = \big[\lambda^2(\mathbf{I}_D + \lambda^2 \nabla s_{kj})\big]^{-1}$, from D JVPs or finite-difference copies, refreshed every `jacobian_refresh` steps.

3. Prior quantities $s_0(g_t), P_0$ and $s_{0,k}(g_t, h_{k,t}), P_{0,k}$ (Section 8).
4. Optional PSD projection of I_{kj} and J_k (Proposition 3).
5. Evaluate (16)–(20) and take the sampler step on all of y.

In the code's current vocabulary, the `hierarchy` argument (shared versus per-observation latents) would become a three-tier index: top-level dimensions, group dimensions with a group id, and local dimensions. Each observation row then carries the group id that tells the sampler which h_k block it shares.

Numerical confirmation. On a linear-Gaussian three-level model ($K = 3$ groups of 4, $d_g = d_h = 2$, $d_\ell = 1$, 23 latent dimensions), we drove a Heun probability-flow ODE with the composed score u from (18)–(20) (400 log-spaced steps, λ from 40 to 10^{-3}, 2×10^4 draws). The ODE was started from the exact diffused law, so that only the score is tested.

sampler score	max |mean err|/sd	sd ratio range	max |corr err|
three-level composed score	0.018	[0.989, 1.007]	0.021
same, group priors $p(g, h_k)^{1-n_k}$ dropped	0.591	[0.948, 2.533]	0.176
Monte Carlo floor (per coordinate / pair)	≈ 0.007		≈ 0.007

The composed-score maxima are taken over 23 means and 253 correlations, so they sit at about three times the per-coordinate Monte Carlo floor, as expected for maxima. The joint posterior, including cross-level correlations (e.g. $\mathrm{corr}(h_0, \ell_{00}) = -0.298$ vs exact -0.294), is recovered. Dropping the mid-level prior corrections biases the means by up to 0.6 posterior sd and inflates some sds by 2.5×.

8 The prior quantities at the mid level

This is the only genuinely new input relative to `gauss_hierarchical`, which needs only (s_0, P_0) on g (closed form for a Gaussian $p(g)$). The group factor needs the diffused *joint* prior score $\nabla \log p_t(g_t, h_{k,t})$ and a backward precision on (g, h).

- **Gaussian** $p(g, h)$ (e.g. $h \mid g \sim \mathcal{N}(Wg+b, \Sigma_h)$ with fixed Σ_h): closed form. $P_{0,k} = \Sigma_{\mathrm{pr},a}^{-1} + \lambda^{-2}\mathbf{I}$ and $s_{0,k} = -(\Sigma_{\mathrm{pr},a} + \lambda^2\mathbf{I})^{-1}(a_{k,t} - \mu_{\mathrm{pr},a})$.
- **Non-Gaussian** $p(g, h)$, typical when g contains the scale of h (funnel): no closed form. The options are (i) a small unconditional score network on (g, h), trained on prior draws only, so it needs no simulator calls; (ii) the same score network with all data masked, if its mask menu includes that pattern; (iii) for `gauss_jacobian`, $P_{0,k}$ from the Tweedie Jacobian of (i)/(ii), exactly like the path precisions. Since all groups share one prior, this is a single model evaluated at K points.
- **Error amplification.** The group prior enters with weight $1 - n_k$, so its score error is multiplied by $n_k - 1$, just as the top prior's error is multiplied by $K - 1$. A poor mid-level prior score therefore hurts most for large groups.

9 From three to N levels

9.1 Nested N-level model

Levels are numbered 1 (top) to N (locals). Nodes at level m are multi-indices $\iota = (i_2, \ldots, i_m)$. The root is $\iota = ()$, and the parent of ι is ι^-, ι with its last index dropped. Let $\mathrm{ch}(\iota)$ be the children of ι and $n_\iota := |\mathrm{ch}(\iota)| \geq 1$. Node ι carries $\theta_\iota \in \mathbb{R}^{d_m}$, and each leaf (level N) has one observation x_ι (several observations per leaf: see Remark 2). The path to ι is $\pi(\iota) = (\theta_{()}, \theta_{(i_2)}, \ldots, \theta_\iota) \in \mathbb{R}^{D_m}$, $D_m = d_1 + \cdots + d_m$, and $\alpha(\iota) = \pi(\iota^-)$ is the path of strict ancestors. The model is

$$p(\theta, \mathbf{x}) = p(\theta_{()}) \prod_{m=2}^{N} \prod_{\iota \in \mathcal{I}_m} p(\theta_\iota \mid \theta_{\alpha(\iota)}) \prod_{\iota \in \mathcal{I}_N} p(x_\iota \mid \theta_{\pi(\iota)}). \tag{22}$$

Three levels: $() = g$, $(k) = h_k$, $(k, j) = \ell_{kj}$. Two levels: $() = g$, $(j) = \ell_j$. The network is trained on full paths $(\theta_{\pi(\iota)}, x_\iota)$, $\iota \in \mathcal{I}_N$, of dimension D_N.

9.2 Step 1 for N levels

Lemma 4 (Counting identity)**.** *For every node ν at level $m < N$, let L_ν be the number of leaves below ν and $\mathcal{T}_\nu$ the internal (non-leaf) nodes of the subtree rooted at ν, including ν. Then*

$$\sum_{\mu \in \mathcal{T}_\nu} (1 - n_\mu) = 1 - L_\nu. \tag{23}$$

Proof. Induction on $N - m$. At $m = N - 1$, $\mathcal{T}_\nu = \{\nu\}$ and $L_\nu = n_\nu$. For $m < N - 1$, $\mathcal{T}_\nu = \{\nu\} \cup \bigcup_{c \in \mathrm{ch}(\nu)} \mathcal{T}_c$, so the sum is $(1 - n_\nu) + \sum_c (1 - L_c) = 1 - n_\nu + n_\nu - L_\nu = 1 - L_\nu$. □

Proposition 5 (Exact N-level factorisation)**.**

$$p(\theta \mid \mathbf{x}) \propto \prod_{m=1}^{N-1} \prod_{\iota \in \mathcal{I}_m} p(\theta_{\pi(\iota)})^{1-n_\iota} \prod_{\iota \in \mathcal{I}_N} p(\theta_{\pi(\iota)} \mid x_\iota), \tag{24}$$

$$\pi_t(\theta_0 \mid y, \mathbf{x}) \propto \prod_{m=1}^{N-1} \prod_{\iota \in \mathcal{I}_m} \pi_t^0(\theta_{\pi(\iota),0} \mid y_{\pi(\iota)})^{1-n_\iota} \prod_{\iota \in \mathcal{I}_N} \pi_t(\theta_{\pi(\iota),0} \mid y_{\pi(\iota)}, x_\iota). \tag{25}$$

Proof. Every factor is a product of the conditional priors along its path; posterior factors also carry their likelihood. The conditional prior of node ν at level m lies on every path through ν. It therefore appears in the L_ν leaf posteriors below ν and in the internal-node prior factors $\mu \in \mathcal{T}_\nu$ (for $m = N$, only in its own leaf factor). Its total exponent is $L_\nu + \sum_{\mu \in \mathcal{T}_\nu} (1 - n_\mu) = 1$ by Lemma 4. Every likelihood appears once, which proves (24). The noise coordinate of θ_ν is split with the same exponents, which sum to one by the same count, and pairing as in Proposition 2 proves (25). □

For $N = 2$ this is $p(g)^{1-n} \prod_j p(g, \ell_j \mid x_j)$. For $N = 3$ it is (3). Each internal node contributes the prior of its whole path with exponent $1 -$ (number of children).

9.3 Steps 2–3 for N levels

Steps 2 and 3 do not use the depth. With Gaussian factors (precision P_ι and network score s_ι on leaf paths, prior precision $P_{0,\iota}$ and score $s_{0,\iota}$ on internal paths),

$$\Lambda = \sum_{m<N} \sum_{\iota \in \mathcal{I}_m} (1 - n_\iota) E_{\pi(\iota)}^\top P_{0,\iota} E_{\pi(\iota)} + \sum_{\iota \in \mathcal{I}_N} E_{\pi(\iota)}^\top P_\iota E_{\pi(\iota)}, \tag{26}$$

and r is the same with each P replaced by Ps. Two nodes are coupled in Λ only if one is an ancestor of the other. Λ is a nested arrowhead of depth N: the arrowhead of level m has, as diagonal blocks, the arrowheads of the level-$(m + 1)$ subtrees.

9.4 Step 4 for N levels: induction over the levels

Definition 1 (Level recursion)**.** *Leaves* ($m = N$): $Q_\iota := P_\iota$ and $\sigma_\iota := s_\iota$ (the network).

Eliminate (any non-root ι, level $m \geq 2$): split Q_ι into the ancestor block $\alpha = \alpha(\iota)$ (dimension D_{m-1}) and the own block ι (dimension d_m), and set

$$S_\iota := [Q_\iota]_{\alpha\alpha} - [Q_\iota]_{\alpha\iota} [Q_\iota]_{\iota\iota}^{-1} [Q_\iota]_{\iota\alpha} = ([Q_\iota^{-1}]_{\alpha\alpha})^{-1}, \qquad G_\iota := -[Q_\iota]_{\iota\iota}^{-1} [Q_\iota]_{\iota\alpha}. \tag{27}$$

Merge (internal ι, level $m < N$): on the path $\pi(\iota)$,

$$Q_\iota := \sum_{c \in \mathrm{ch}(\iota)} S_c + (1 - n_\iota) P_{0,\iota}, \qquad \sigma_\iota := Q_\iota^{-1} \Big[\sum_{c \in \mathrm{ch}(\iota)} S_c \, \sigma_c^{\alpha(c)} + (1 - n_\iota) P_{0,\iota} \, s_{0,\iota} \Big], \tag{28}$$

where $\sigma_c^{\alpha(c)}$ is the restriction of σ_c to $\alpha(c) = \pi(\iota)$.

Theorem 6 (N-level composition rule)**.** *Compute Definition 1 for $m = N, N-1, \dots, 1$. Then set*

$$u^{()} = \sigma_{()}, \qquad u^{\iota} = \sigma_{\iota}^{\iota} + G_{\iota}(u^{\alpha(\iota)} - \sigma_{\iota}^{\alpha(\iota)}) \quad (m = 2, \dots, N, \ \textit{top-down}), \tag{29}$$

where $u^{\alpha(\iota)}$ is the already computed composed score on the ancestors. If all pivots $[Q_\iota]_{\iota\iota}$ ($m \geq 2$) and $Q_{()}$ are positive definite, u is the unique solution of $\Lambda u = r$ with (26), i.e. the composed score of all *latents at all levels.*

Proof. We prove, by downward induction on $m = N, \dots, 1$:

Claim(m). After all nodes at levels $> m$ have been eliminated, the reduced system is the system (26) of the *truncated m-level model*, in which each level-m node ι carries the single merged factor $(Q_\iota, Q_\iota\sigma_\iota)$ on $\pi(\iota)$ in place of its subtree, together with the prior factors of the levels $< m$.

Base $m = N$: nothing has been eliminated, and the leaf factors are $(P_\iota, P_\iota s_\iota) = (Q_\iota, Q_\iota\sigma_\iota)$.

Step $m \to m-1$, with $m \geq 2$. Fix $\iota \in \mathcal{I}_m$. By Claim(m), block θ_ι appears only in its merged factor: all other remaining factors live on paths of other level-m nodes or of levels $< m$, and none contains θ_ι. Its row reads $[Q_\iota]_{\iota\alpha}u^{\alpha} + [Q_\iota]_{\iota\iota}u^{\iota} = [Q_\iota]_{\iota\alpha}\sigma_\iota^{\alpha} + [Q_\iota]_{\iota\iota}\sigma_\iota^{\iota}$, which is (29). Substituting it into the factor's contribution to the α rows, $[Q_\iota]_{\alpha\alpha}u^{\alpha} + [Q_\iota]_{\alpha\iota}u^{\iota}$ against $[Q_\iota]_{\alpha\alpha}\sigma_\iota^{\alpha} + [Q_\iota]_{\alpha\iota}\sigma_\iota^{\iota}$, cancels every σ_ι^{ι} term and leaves $S_\iota u^{\alpha} = S_\iota\sigma_\iota^{\alpha}$. Hence ι is replaced by the message $(S_\iota, S_\iota\sigma_\iota^{\alpha})$ on $\alpha(\iota) = \pi(\iota^{-})$. For a level-$(m-1)$ node μ, the remaining factors with scope exactly $\pi(\mu)$ are the n_μ messages of its children and its prior factor $(1-n_\mu)P_{0,\mu}$. No other factor contains θ_μ, since prior factors of levels $< m-1$ live on shorter paths. Summing them gives $(Q_\mu, Q_\mu\sigma_\mu)$ of (28), which is Claim($m-1$).

End $m = 1$: the root's factor is $(Q_{()}, Q_{()}\sigma_{()})$, so $u^{()} = \sigma_{()}$. Back-substitution in reverse elimination order gives (29). The passes form a block $LDL^{\top}$ factorisation with the stated pivots, which proves uniqueness and the positive-definiteness criterion. □

N-level composed score

Upward (levels $N \to 1$): at the leaves, $(Q, \sigma) = (P, s)$ from the network. At each node, Schur-reduce onto the ancestors, $S_\iota = ([Q_\iota^{-1}]_{\alpha\alpha})^{-1}$, and send $(S_\iota, \sigma_\iota^{\alpha})$ to the parent. The parent merges $Q = \sum_c S_c + (1-n)P_0$ and $\sigma = Q^{-1}[\sum_c S_c\sigma_c^{\alpha} + (1-n)P_0 s_0]$.

Downward (levels $1 \to N$): $u^{()} = \sigma_{()}$, then $u^{\iota} = \sigma_\iota^{\iota} + G_\iota(u^{\alpha(\iota)} - \sigma_\iota^{\alpha(\iota)})$.

Reading the theorem as "two levels, recursively". Claim(m) says that *a level-m subtree behaves exactly like one observation of an m-level model.* Its "network score" is σ_ι, the composed score of the subtree's own tall posterior on its path, and its "precision" is Q_ι. Every merge (28) is literally the `gauss_hierarchical` global rule. Every elimination (27) together with the back-substitution (29) is literally the `gauss_hierarchical` local rule ($S \leftrightarrow S_j$, $G \leftrightarrow R_j$). The induction can also be run top-down. Split the N-level model at the root into a two-level model whose "observations" are the level-2 subtrees. Each subtree is an $(N-1)$-level model with global block $(\theta_{()}, \theta_{(i_2)})$ and prior $p(\theta_{()}, \theta_{(i_2)})$. Its tall posterior uses exactly the factors of (24) that lie in the subtree, because Lemma 4 makes their exponents cover the subtree's path once. The root combines the subtrees by the two-level rule. This gives the same (Q, σ) and the same answer.

Reductions. $N = 2$ is `gauss_hierarchical` (30) below. $N = 3$ is the boxed rule of Section 5, with (S_{kj}, K_{kj}) from the leaf eliminations, (Q_k, σ_k) from the group merges, (T_k, G_k) from the group eliminations, and the root merge giving (18):

$$N = 2: \quad u^{g} = \Big[\sum_j S_j + (1-n)P_0\Big]^{-1}\Big[\sum_j S_j s_j^{g} + (1-n)P_0 s_0\Big], \quad u^{\ell_j} = s_j^{\ell} + R_j(u^{g} - s_j^{g}). \tag{30}$$

The local correction at depth N. Expanding (29) along the path, with $G_\iota = (G_\iota^{(1)}, \dots, G_\iota^{(m-1)})$ split by ancestor level,

$$u^\iota = \sigma_\iota^\iota + \sum_{a=1}^{m-1} G_\iota^{(a)} (u^{(a)} - \sigma_\iota^{(a)}).$$

A node at level m is corrected by the composed-minus-own discrepancy at *every* ancestor level, each weighted by its regression coefficient under the node's subtree kernel. For leaves, $\sigma_\iota = s_\iota$ is the raw network score, as in (20).

Well-posedness and safeguard. In information form, $Q_\iota = P_{0,\iota} + \sum_{c \in \mathrm{ch}(\iota)} (S_c - P_{0,\iota})$. If every increment is PSD (true for linear-Gaussian models, by the conditioning argument of Section 6), all pivots are positive definite. The PSD projection is applied once per internal node.

Cost. $\sum_{m=1}^{N} |\mathcal{I}_m| \, \mathcal{O}(D_m^3)$, i.e. linear in the number of nodes and cubic in the path dimension. A full Λ solve would cost $\mathcal{O}((\sum_m |\mathcal{I}_m| d_m)^3)$.

What has to be supplied at depth N.

- Network: the joint score on full root-to-leaf paths of dimension D_N, and the path precision P_ι (pilot covariance for `gauss_hierarchical`, D_N JVPs/FD copies for `gauss_jacobian`).
- Priors: the diffused score and backward precision of $p(\theta_{\pi(\iota)})$ for every internal level $m = 1, \dots, N-1$ with $n_\iota \neq 1$. For exchangeable designs this means one prior model per level.

Remark 2 (Ragged trees, several observations per leaf, data at inner levels). None of the proofs uses equal branching. If a node ι also has o_ι observations attached directly (several trials per local, or data that depend only on (g, h_k)), each one enters the merge (28) like a child, with (S_c, σ_c) replaced by the network's (P_o, s_o) on $\pi(\iota)$. The prior exponent becomes $1 - n_\iota - o_\iota$. The companion note `multilevel_composition.tex` gives the general tree and DAG form.

9.5 Numerical confirmation

The level recursion of Theorem 6 and a literal transcription of the boxed three-level rule were compared with the exact diffused tall-data score of random linear-Gaussian hierarchies, at $\lambda \in [10^{-3}, 10]$ (`check_three_level_joint_sampling.py`).

hierarchy	latent dim	observations	max rel. error
$N = 2$, branching 6	8	6	1.2×10^{-15}
$N = 3$, branching 3×3	17	9	1.3×10^{-15}
$N = 3$, branching 3×4, explicit boxed rule	23	12	1.4×10^{-15}
$N = 4$, branching $2 \times 3 \times 2$	23	12	1.3×10^{-15}
$N = 5$, branching $2 \times 2 \times 2 \times 2$	48	16	3.8×10^{-15}
$N = 3$, group priors $p(g, h_k)^{1-n_k}$ dropped	23	12	0.011–0.82

The ablation error is 0.011 at $\lambda = 10$ and grows to 0.82 at $\lambda = 0.01$.

10 Summary

1. The three-level hierarchy factorises exactly as $p(g)^{1-K} \prod_k p(g, h_k)^{1-n_k} \prod_{kj} p(g, h_k, \ell_{kj} \mid x_{kj})$, both for clean densities and for diffusion backward kernels.
2. Under `gauss_hierarchical`'s single Gaussian-kernel approximation, the composed score of *all* latents solves a nested arrowhead system, which three elimination passes solve in closed form (16)–(20). Only network path scores enter, and both the local and the mid-level own-scores cancel from the levels above. Linear-Gaussian models are exact.

3. The rule yields a joint score for top globals, mid globals and locals, and therefore joint posterior sampling with any reverse-diffusion sampler (verified to Monte Carlo noise).

4. The N-level rule follows by induction: merge $(\sum S + (1 - n)P_0)$ and eliminate (Schur, regression) once per level upwards, then cascade conditional-mean corrections downwards. It is `gauss_hierarchical` applied recursively, with each subtree acting as one observation, at cost linear in the number of nodes.

5. The new requirement is a diffused prior score and precision on every intermediate path, e.g. (g, h), weighted by $1 - n$. In partial pooling this prior is not Gaussian.